

Deep Learning for AI-Powered Job Recommendations

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Abstract

We investigate multi-modal deep learning architectures for job recommendation systems by combining structured attributes (e.g., experience level, salary quantile) with unstructured textual information (e.g., job titles, descriptions). We evaluate three models: a baseline BERT-Base model trained only on text, a Fusion model using cross-attention to dynamically integrate structured and textual features, and a Fusion-Concatenation model employing simple feature concatenation. Using a large LinkedIn job postings dataset, we find that the Fusion model achieves the highest macro F1-score (0.815), but the BERT-Base model performs nearly as well (0.811) with far lower complexity. While cross-attention provides modest gains, most structured signals are already embedded in rich textual data. Our results suggest that simpler text-only models offer an attractive tradeoff between performance and efficiency for real-world deployment. We also highlight the challenges in classifying medium salary ranges, where misclassifications persist across models, indicating potential limitations in data granularity or salary definition clarity.

1. Introduction/Background/Motivation

The task of providing personalized job recommendations is critical for improving the efficiency of the labor market by matching candidates with optimal career opportunities. Traditional job recommendation systems typically rely on collaborative filtering approaches, or content-based filtering systems. Collaborative filtering methods match candidates to jobs based on similar users' preferences but struggle with the 'cold start' problem for new users and jobs without interaction history. Content-based methods primarily use keyword matching between resumes and job descriptions, often missing semantic relationships and contextual nuances in language. Both approaches heavily depend on structured data (location, years of experience, job titles) and fail to fully leverage the rich contextual information embedded in unstructured text. This creates a significant gap in recommendation quality, particularly when matching can-

didates to positions requiring specialized skills or when job descriptions contain domain-specific terminology that extends beyond simple keyword matching. Recent advances in multi-modal deep learning have opened new avenues for integrating both structured data and unstructured textual information, such as job titles, required skills, and descriptions, to generate more precise and individualized recommendations. Multi-modal approaches that combine structured attributes with textual data have already demonstrated improved predictive capabilities across domains such as healthcare and finance (1), further motivating their application to job recommendation tasks.

In this project, we explore whether multi-modal deep learning architectures that combine structured attributes with unstructured textual information can enhance job recommendation accuracy compared to unimodal approaches. We investigate if a cross-attention fusion mechanism, which dynamically integrates structured categorical features with contextual embeddings from job descriptions, can capture additional signal beyond what is present in text alone. This research question directly addresses current recommendation systems' limitations by examining, if the integration of multiple data types offers meaningful improvements or if the rich semantic information in job descriptions already encodes most of the necessary signals for accurate recommendations. This investigation could reveal important insights about model complexity trade-offs in practical recommendation systems.

Through extensive experimentation, we find that while cross-attention offers a slight performance improvement, the fine-tuned BERT-Base model, despite its simplicity, achieves nearly equivalent predictive accuracy. This finding highlights important trade-offs between model complexity and practical deployment considerations.

1.1. Why It Matters

The impact of improved job recommendation systems extends beyond simple market efficiency. For job seekers, our approach could reduce the average job search duration, currently 5 months according to recent labor statistics, by presenting more relevant opportunities earlier in

their search, potentially decreasing periods of unemployment and associated financial stress. Employers benefit through reduced time-to-hire metrics and lower recruitment costs, which currently average \$4,700 per position according to industry estimates. Higher quality matches also lead to improved employee retention and performance, as candidates are more likely to thrive in roles that truly align with their skills and career aspirations. On a macroeconomic level, better job matching addresses skill gaps in high demand sectors, potentially increasing workforce productivity and reducing structural unemployment. This is particularly valuable in today’s rapidly evolving job market, where technological disruption continuously transforms skill requirements across industries.

2. Dataset and Exploratory Data Analysis

We utilized the publicly available Kaggle dataset *LinkedIn Job Postings (2023–2024)* (4) comprising over 124,000 job postings listed across 2023 and 2024. The dataset spans diverse industries, experience levels, work types, and geographic locations. It includes both structured features (e.g., experience level, salary quantile, work type, and location) and unstructured textual information (e.g., job titles, required skills, and job descriptions). Figures 1, 2, and 3 illustrate key exploratory data analysis (EDA) results.

Figure 1 shows the distribution of experience levels in the dataset. The majority of postings target Mid-Senior level professionals, followed by Entry-level roles and postings with an Unknown experience level. Associate positions make up a smaller portion, while Director, Executive, and Internship roles are relatively rare. This distribution suggests a market skewed toward more experienced candidates, an important factor influencing model training and prediction performance.

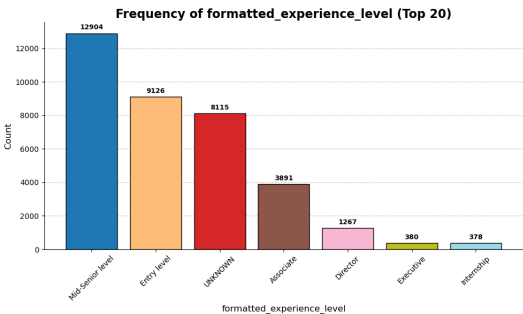


Figure 1. Distribution of Experience Levels in the Dataset.

Figure 2 highlights the 20 most common job titles. Positions such as Senior Accountant, Administrative Assistant, Project Manager, and Staff Accountant appear frequently, reflecting high demand for finance, administrative, and project management roles during this period. Many job titles show relatively modest frequencies, indicating a

long-tailed distribution of role specificity that could benefit from the semantic capabilities of pre-trained language models like BERT. Together, these exploratory insights validate the multimodal design of our approach: while structured attributes like experience level and work type are important, the textual richness captured by job titles and descriptions offers deeper granularity essential for personalized job matching.

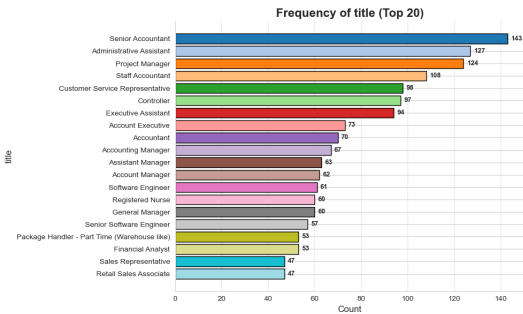


Figure 2. Top 20 Most Frequent Job Titles.

Figure 3 further explores the relationship between experience level and annual salary. As shown, there is a clear positive correlation between experience level and median salary. Internships and Entry-level positions offer the lowest median salaries (\$47,500 and \$51,806 respectively), while Director and Executive roles command substantially higher medians (\$165,000 and \$180,000 respectively). Mid-Senior level positions, which dominate the dataset in volume, have a median salary of approximately \$105,000. The boxplot also highlights the substantial variability in salaries within each experience category, particularly for Mid-Senior and Director levels. The presence of outliers and the wide interquartile ranges indicate that job titles and industries can heavily influence salary outcomes even within the same experience bracket. These findings reinforce the need for models that can capture fine-grained textual and structured features rather than relying solely on high-level attributes like experience level.

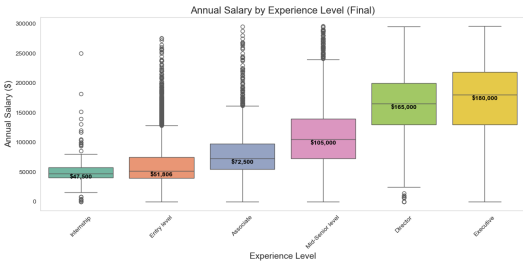


Figure 3. Annual Salary Distribution by Experience Level.

3. Approach

We explored three deep learning architectures to evaluate the efficacy of multimodal fusion strategies for the job recommendation task. The first model, the **Baseline Model (BERT-Base)**, utilized a pre-trained BERT model (bert-base-uncased) fine-tuned exclusively on the textual fields of each job posting, including job title, required skills, and job description. This model served as a strong unimodal baseline without incorporating any structured data. The second architecture, the **Fusion Model (Cross-Attention)**, introduced a multimodal approach by integrating structured attribute embeddings—specifically experience level, industry, location, and salary quantile—generated by a multi-layer perceptron (MLP) with BERT-based textual embeddings through cross-attention layers. Inspired by prior studies on multimodal fusion strategies, particularly the use of cross-attention to integrate structured and unstructured modalities (5), we designed a fusion model that dynamically attends to structured attributes during inference. This dynamic fusion mechanism allowed the model to condition its predictions jointly on textual and structured information, attending to relevant attributes during inference. The integration of structured categorical attributes into contextual embeddings follows the principle that contextualized representations improve model robustness and predictive performance (3). The third model, the **Fusion-Concatenation Model**, employed a simpler fusion strategy where structured embeddings were concatenated with BERT-generated textual embeddings and passed through additional fully connected layers. This model evaluated whether basic integration without dynamic attention could achieve performance competitive with more complex architectures. Across all models, we used consistent hyperparameters: a learning rate of 2×10^{-5} , a batch size of 64, and a maximum sequence length of 256 tokens. Structured embeddings were set to a dimensionality of 64, and the hidden dimension for fully connected layers was set to 128. All BERT layers were fully fine-tuned during training. Models were trained for 4 epochs using a cross-entropy loss function optimized with the AdamW optimizer, and our GitHub repository can be found here: (2).

3.1. Implementation Specifics

Our implementation used PyTorch and the Hugging Face Transformers library to create all three models. For the cross-attention mechanism in our Fusion model, we employed the native PyTorch MultiheadAttention with 8 attention heads, where structured attribute embeddings served as queries while BERT’s contextual representations functioned as keys and values. This approach allowed the model to dynamically weight textual features based on structured attribute relevance. The implementation followed the sequence: (1) encode text through BERT, (2) encode cate-

gorical attributes through embedding layers followed by an MLP, (3) perform cross-modal fusion either through attention or concatenation, and (4) feed the fused representation through a classification head.

3.2. Design Justification

Our hyperparameter choices were informed by practical considerations and common practices in multi-modal networks. We selected a structured embedding dimension of 64 to balance expressiveness and computational efficiency, large enough to capture attribute variations but small enough to prevent overfitting given our dataset size. Similarly, the hidden dimension of 128 for fully connected layers provided sufficient capacity for feature transformation while maintaining model parsimony. The learning rate of 2×10^{-5} was chosen specifically to accommodate BERT fine-tuning requirements, as larger learning rates could destabilize pre-trained weights. Our batch size of 64 struck a balance between computation efficiency and stochastic gradient benefits.

3.3. Problems and Iterations

We anticipated several challenges at the start of the project. First, we expected potential class imbalance issues given the natural salary distribution skew in the job markets. This was addressed through stratified sampling during train/validation/test splits to maintain consistent class proportions. Second, we anticipated data quality challenges that materialized as the dataset contained numerous missing values and inconsistent formatting in location fields, unstructured job descriptions, and outlier salaries. Our cleaning pipeline handled all of this through a combination of several cleaning techniques.

The initial training attempts revealed optimization difficulties with the Fusion Concatenation model, which showed higher initial validation loss and slower convergence than expected. We addressed this by implementing a two phase training approach with a head phase warmup period, allowing the classification head to stabilize before fine tuning the entire network. Computational constraints also emerged when processing lengthy job descriptions, many exceeding 1,000 tokens, forcing us to truncate to 256 tokens and implement efficient batching. Our first cross-attention implementation suffered from gradient instability, requiring the addition of layer normalization and adjusting the attention temperature to achieve stable training.

3.4. Contribution and Approach Differentiation

Although our work builds upon existing cross-attention mechanisms, our approach introduces a novel application of multi-modal fusion specifically tailored to job recommendation systems. Unlike previous work that primarily focused on image-text or speech-text pairs, our model bridges

structured categorical attributes with unstructured job descriptions, a combination rarely explored in employment matching literature. Our contribution lies in demonstrating how different fusion strategies perform in this specific domain, revealing that the marginal benefit of sophisticated fusion mechanisms may be limited when textual representations already implicitly encode much of the structured information, an insight relevant to practitioners weighing model complexity against performance gains.

4. Experiments and Results

To evaluate our models, we used multiple performance metrics, including validation loss, precision, recall, F1-score, and class-specific F1-scores. Confusion matrices further provided insight into model mis-classification patterns. Figure 4 shows the validation loss over epochs for each model. The BERT-Base model consistently achieved the lowest validation loss across training, indicating rapid convergence and strong generalization. The Fusion model (cross-attention) also demonstrated stable convergence, ultimately achieving slightly lower loss than the Fusion-Concatenation model.

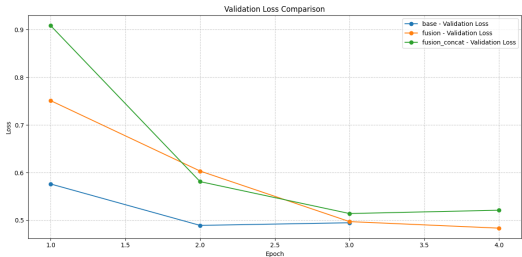


Figure 4. Validation Loss Comparison Across Models.

Macro-averaged performance metrics are summarized in Figure 5. While all models achieved relatively strong performance, the Fusion model achieved the highest macro F1-score (0.815), followed closely by the BERT-Base model (0.811). The Fusion-Concatenation model lagged slightly behind with a macro F1-score of 0.792. Precision and recall metrics show a similar pattern, suggesting that the cross-attention mechanism provided a modest but measurable benefit.

Class-specific F1 scores are shown in Figure 6. The Fusion model slightly outperformed the other models across all salary ranges (low, medium, high). Notably, the Medium salary class consistently exhibited lower F1 scores across models, indicating a more challenging classification boundary compared to Low and High categories.

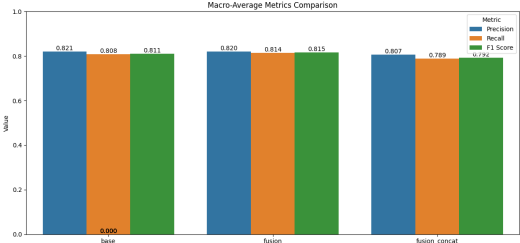


Figure 5. Macro-Average Metrics (Precision, Recall, F1 Score) Comparison.

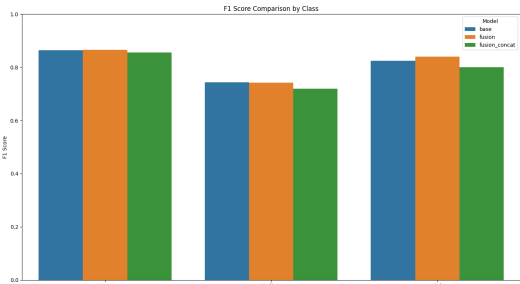


Figure 6. F1 Score Comparison by Class (Low, Medium, High).

Finally, confusion matrix analysis further highlights performance differences. Figure 7 presents the normalized confusion matrix for the best-performing Fusion model. Most misclassifications occurred between adjacent salary ranges, particularly for the Medium class, confirming that Medium salaries represent a "gray area" that models struggled to classify accurately.

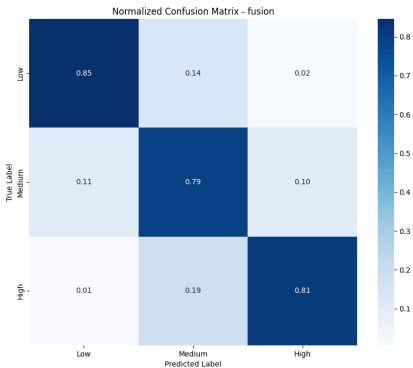


Figure 7. Normalized Confusion Matrix for the Fusion Model (Cross-Attention).

Overall, results support our hypothesis that cross-attention fusion can offer modest improvements over simpler baselines, though the BERT-Base model remained highly competitive.

5. Discussion

Our experiments reveal that the Fusion model utilizing cross-attention slightly outperforms the BERT-Base model in terms of macro-average F1-score and precision. This modest boost in performance can be attributed to the model’s ability to dynamically integrate structured features with textual embeddings. The cross-attention mechanism enables the model to attend to relevant structured attributes (e.g., experience level, industry, location) when interpreting job descriptions, thereby improving contextual understanding and reducing misclassifications across salary quantiles. However, despite this improvement, the BERT-Base model remains highly competitive. Its strong performance is likely due to the rich semantic information already embedded within pre-trained language models such as BERT. Job titles, required skills, and job descriptions often implicitly encode indicators like seniority, industry, and even geographic hints, lessening the marginal utility of structured attributes. Consequently, the simpler BERT-Base model, requiring fewer parameters, reduced model complexity, and faster training, offers a highly attractive trade-off between accuracy and operational efficiency. Similar to findings in multimodal deep learning research (5), our experiments revealed that simpler uni-modal architectures leveraging rich textual information can perform competitively against more complex fusion models. A closer examination of the confusion matrices indicates that all models, including the Fusion variant, struggled most with the “Medium” salary quantile classification. Misclassifications frequently occur between adjacent salary classes (Low and Medium, or Medium and High), suggesting substantial overlap in the features that define these groups. This ambiguity likely arises because mid-range salaries encompass a broader and less sharply delineated set of roles, complicating classification. While cross-attention fusion can yield slight performance gains, the results underscore an important insight: richer data representations and better feature separability may ultimately provide greater returns than increasing model complexity alone.

5.1. Hypothesis Evaluation

Our results provided only partial support for our initial hypothesis that multi-modal fusion would enhance job recommendation accuracy, while the cross-attention fusion model did achieve the highest macro F1-score (0.815) compared to BERT-Base (0.811) and Fusion-Concatenation (0.792), the performance gain was marginal rather than substantial. This suggests that the semantic information contained in job descriptions and titles already implicitly encodes much of the structured attribute information we tried to integrate explicitly. The small performance difference between models indicates that complex fusion architectures may offer diminishing returns in this specific domain, chal-

lenging our assumption about the complementary nature of structured and unstructured data for job salary prediction.

5.2. Qualitative Analysis of Examples

Analyzing specific misclassifications revealed interesting patterns across models. For instance, a ‘Secret Shopper’ position with a high actual salary was consistently misclassified as low by all three models. Examining this example (visible in all three models’ error analysis), we found the job description emphasized data collection responsibilities while downplaying management aspects that would typically signal higher compensation. Similarly, technical roles like ‘Remote Software Engineer (Python)’ with low actual salaries were frequently misclassified as high across all models, likely due to the presence of technical keywords (‘AWS’, ‘Python’, ‘Docker’) that commonly signal higher compensation. These consistent mis-classification patterns across models suggest that certain job types contain inherently ambiguous salary signals that even sophisticated fusion mechanisms struggle to resolve.

5.3. Why Cross-Attention Didn’t Provide Substantial Gains

The modest improvement from cross-attention can be attributed to several factors revealed in our error analysis. First, examining the mis-classification counts shows that while the fusion model reduced total errors (1,005) compared to the BERT-Base model (1,037), the improvement was marginal, approximately 3%. Second, the misclassification patterns remained largely consistent across models, all struggled most with the Medium salary class, with most errors involving adjacent salary categories. For example, the High-to-Medium misclassification was the most common error type across all models (405 instances in BERT-Base, 336 in Fusion, 502 in Fusion-Concatenation). This pattern suggests a fundamental ambiguity in the salary boundaries that additional modalities couldn’t fully resolve. Finally, the simple concatenation approach performed noticeably worse (21.02% error rate) than either the attention-based fusion (18.58%) or the pure textual approach (19.17%), indicating that naive integration of modalities can actually degrade performance, likely by introducing noise that disrupts the strong textual signals.

5.4. What Worked and What Didn’t

Our approach succeeded in demonstrating that a relatively simple BERT-Base model could achieve strong performance on salary prediction without complex multi-modal integration. The cross-attention mechanism showed modest improvements, particularly in reducing High-to-Medium misclassifications (from 405 instances in BERT-Base to 336 in Fusion). However, our approach fell short in several key areas. First, the significant computational

overhead of the fusion models didn't translate to proportional performance gains. Second, despite our integration of structured attributes, all models still struggled with Medium salary predictions. It's a confirmation that categorical boundaries in salary prediction remain inherently challenging regardless of model architecture. Third, the simple concatenation approach degraded performance, suggesting that not all multi-modal integration strategies are beneficial. These findings suggest that for production systems in job recommendation, the marginal gains from complex fusion may not justify the increased implementation complexity and computational costs.

6. Conclusion

In this project, we investigated multi-modal deep learning approaches for personalized job recommendations by combining structured attributes with unstructured textual job descriptions. Our experiments demonstrate that while a cross-attention fusion model provides a marginal performance gain (F1-score of 0.815 versus 0.811), the baseline BERT-Base model achieves nearly equivalent predictive accuracy with a simpler, more efficient architecture. This finding reveals an important insight: the rich semantic information inherently captured by pre-trained language models already encodes much of the signal present in structured data fields, challenging the assumption that complex fusion mechanisms necessarily yield substantial benefits in job recommendation systems. The tradeoff between marginal performance improvements and increased computational complexity makes the BERT-Base model more attractive for practical deployment scenarios where efficiency and maintainability are important considerations. Our analysis of mis-classification patterns across models further revealed consistent errors in categorizing Medium salary jobs, suggesting inherent ambiguity in the boundaries between salary categories that persists regardless of model architecture. This points to fundamental challenges in the job recommendation domain that may benefit more from improved data representation and feature engineering than from architectural complexity alone. For organizations building job recommendation systems, our findings suggest that investing in high-quality job description text and optimizing simpler models may provide better returns than implementing complex multi-modal architectures with only marginal performance benefits. Future work could explore integrating additional contextual signals beyond what we investigated, such as company reputation scores, industry growth forecasts, or geographic economic indicators. There's also potential to enhance recommendation quality through multi-task learning approaches that simultaneously predict multiple job attributes (salary, experience requirements, and growth potential), potentially creating richer shared representations. As labor markets continue to evolve with chang-

ing skill requirements and work arrangements, recommendation systems that can efficiently balance accuracy and computational efficiency will become increasingly valuable tools for both job seekers and employers navigating the complex employment landscape. (1).

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Student Name	Contributed Aspects	Details
Eddie Capone	EDA and Reporting	Conducted exploratory data analysis (EDA), wrote the report, and submitted the final project.
Pradeep Divile	Baseline Model Training	Trained BERT-Base and computed performance metrics.
Gavin Williams	Fusion Model Training	Trained the fusion models and computed performance metrics.
Capone, Divile, Williams	Final Review	Reviewed the rough draft and finalized the report collaboratively.

Table 1. Contributions of team members.